Supplementary Information
Transcriptome Universal Single-isoform COntrol (TUSCO): a benchmarking framework for transcriptome reconstruction quality

Tianyuan Liu[1,2], Alejandro Paniagua[1,2], Fabian Jetzinger[1,2,3], Luis Ferrández-Peral[1,5], Adam Frankish[4], Ana Conesa[1*]

[1]Institute for Integrative Systems Biology (I2SysBio), Spanish National Research Council (CSIC), Paterna 46980, Spain.
[2]Department of Computer Science, University of Valencia, Spain.
[3]BioBam Bioinformatics S.L., Valencia, 46006, Spain.
[4]European Molecular Biology Laboratory, European Bioinformatics Institute, Wellcome Genome Campus, Hinxton, Cambridge CB10 1SA, United Kingdom.
[5]Current address: Biozentrum, University of Basel, Basel CH-4056, Switzerland.

Contents

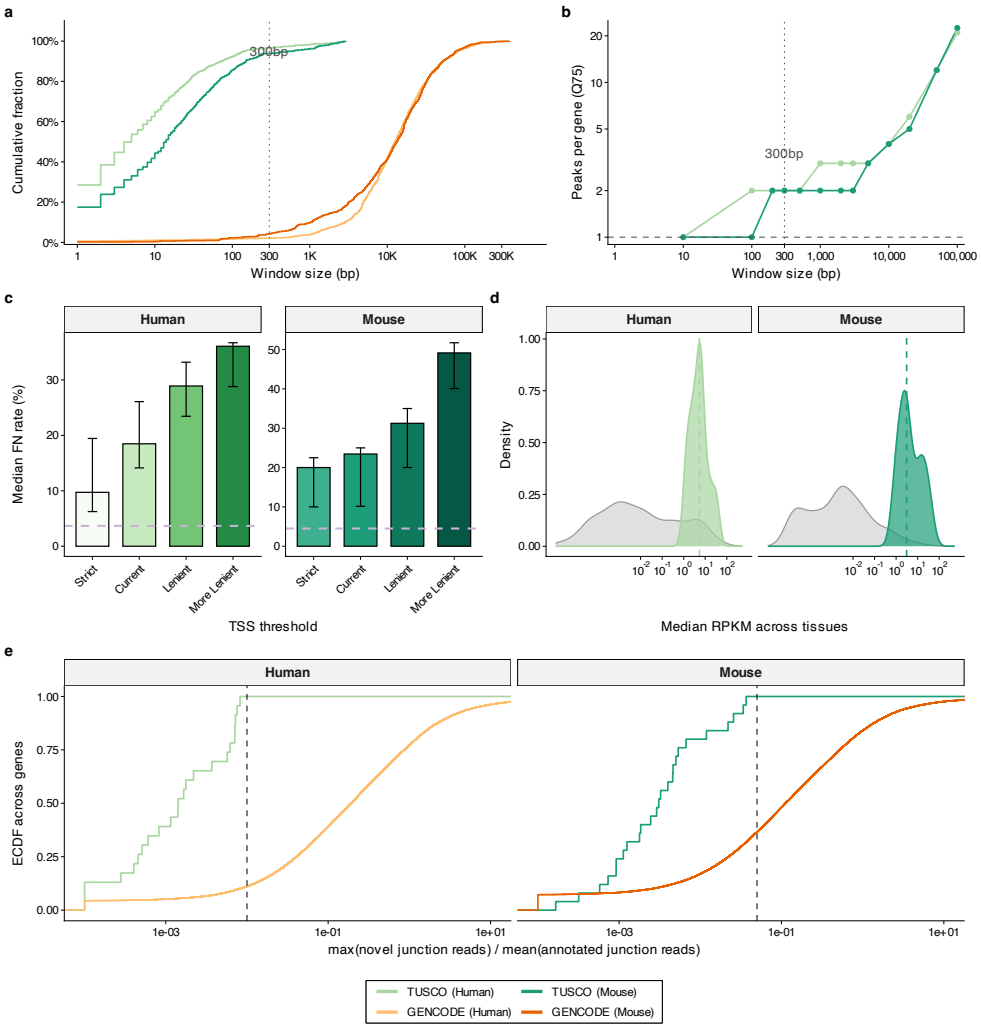
1 Supplementary Figures **2**

2 Supplementary Tables **10**

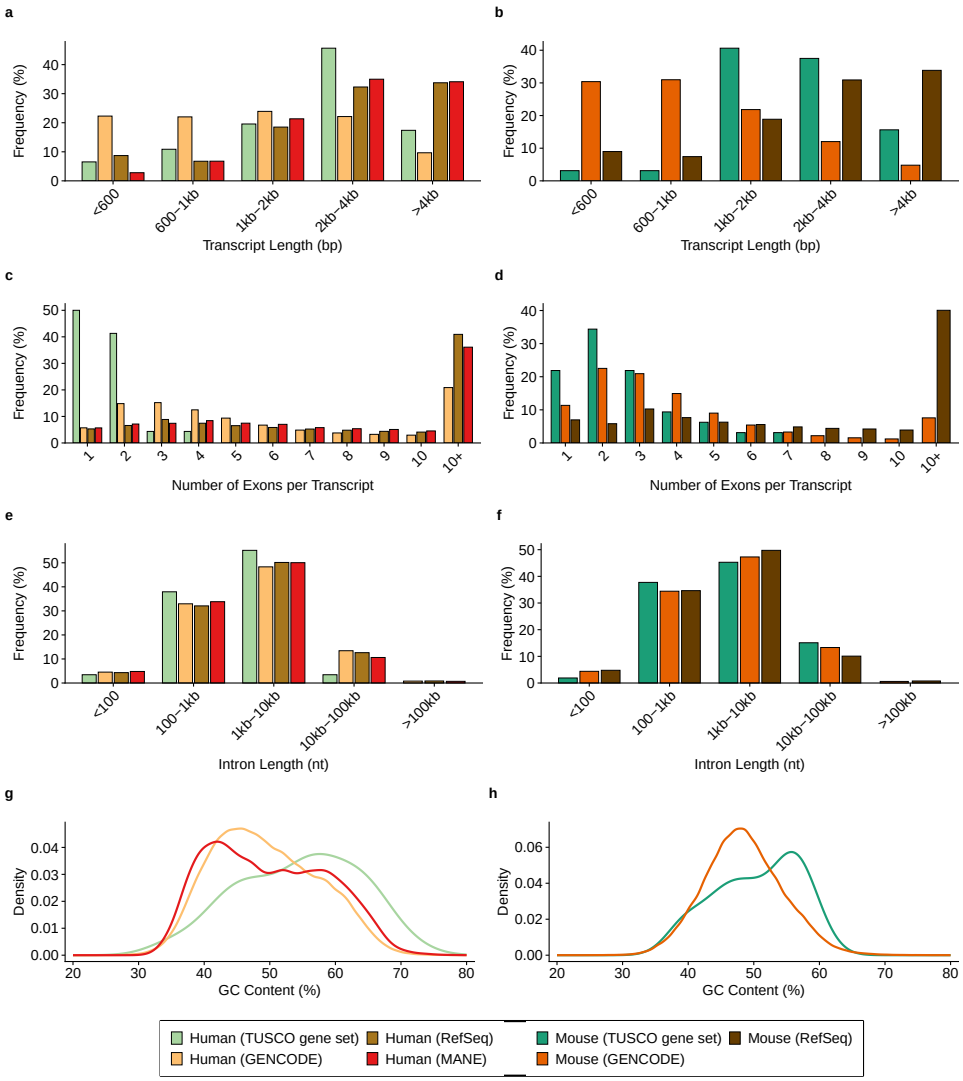
3 Supplementary Notes: TUSCO False Positive Validation **12**

1 Supplementary Figures

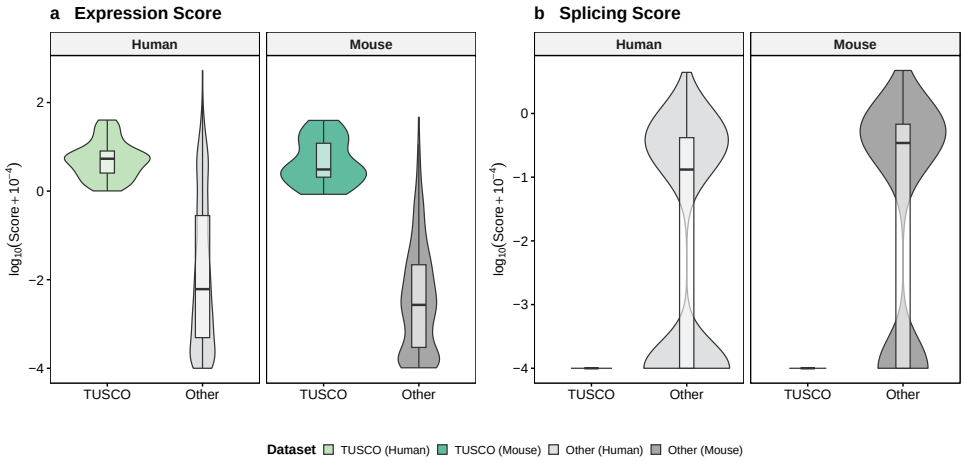
This section compiles visual summaries referenced throughout the manuscript. Supplementary Fig. 1 presents parameter selection sensitivity analyses for the TUSCO benchmarking framework. Supplementary Fig. 2 contrasts fundamental transcript properties between the TUSCO gene set and canonical annotations in human and mouse, while Supplementary Fig. 3 benchmarks the expression and splicing invariance of genes in the TUSCO gene set against other single-isoform loci. Supplementary Fig. 4 shows false-negative rates from TUSCO evaluation grouped by sequencing platform and method. Supplementary Fig. 5 compares sensitivity, false-negative rate, and false discovery rate between TUSCO evaluation and MAJIQ-L across all pipelines. Supplementary Fig. 6 examines per-read coverage distributions for transcripts in the TUSCO gene set, comparing TP and PTP calls and stratified by reference transcript exonic length. Supplementary Fig. 7 evaluates how SIRV-based concordance metrics respond to replicate number, contextualizing the stability of spike-in controls relative to endogenous benchmarks.



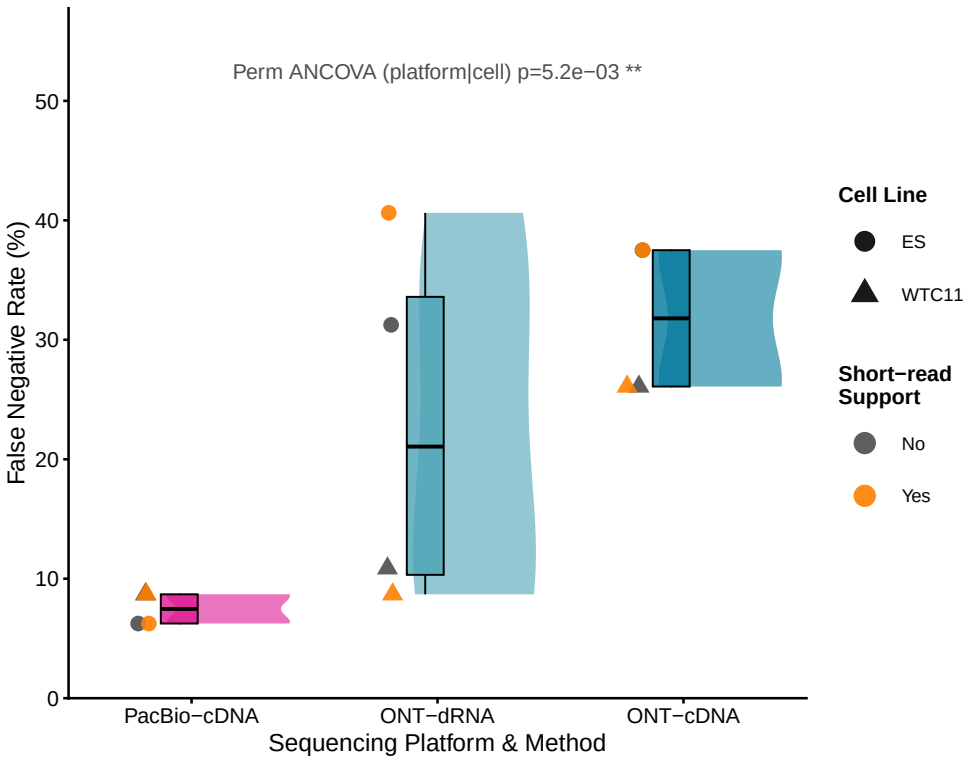
Supplementary Figure 1: Parameter selection sensitivity analysis for the TUSCO benchmarking framework. a, TSS/CAGE window size validation showing cumulative fraction of genes whose annotated TSS (from refTSS database) falls within varying window sizes; "TUSCO" curves show within-gene sensitivity, while "GENCODE" curves show inter-gene collision rates. At ±300 bp, human data show 96.6% sensitivity against only 2.1% inter-gene collision. **b**, Distribution of refTSS peak counts per gene at varying window sizes; values above 1.0 indicate potential ambiguity from multiple TSS peaks. **c**, Expression threshold sweep comparing TUSCO false-negative rates (from LRGASP evaluation) against SIRV baselines (dashed line) across four threshold configurations (Strict to More Lenient) for human and mouse. The "Current" configuration balances gene count against a sharp rise in false negatives under more lenient settings. **d**, Distribution of AlphaGenome-predicted tissue-median RPKM for genes in the TUSCO gene set (colored) versus other single-isoform genes (gray); dashed lines indicate medians for genes in the TUSCO gene set, demonstrating that they occupy the high-expression tail (90th–97th percentile). **e**, Novel splice junction threshold validation using AlphaGenome-predicted junction read counts, showing ECDF of the ratio max(novel junction reads)/$\mu_{\text{annotated}}$ for genes in the TUSCO gene set versus other single-isoform genes. Genes in the TUSCO gene set cluster well below the chosen α threshold (dashed lines; 0.01 for human, 0.05 for mouse), whereas excluded genes show markedly higher ratios.



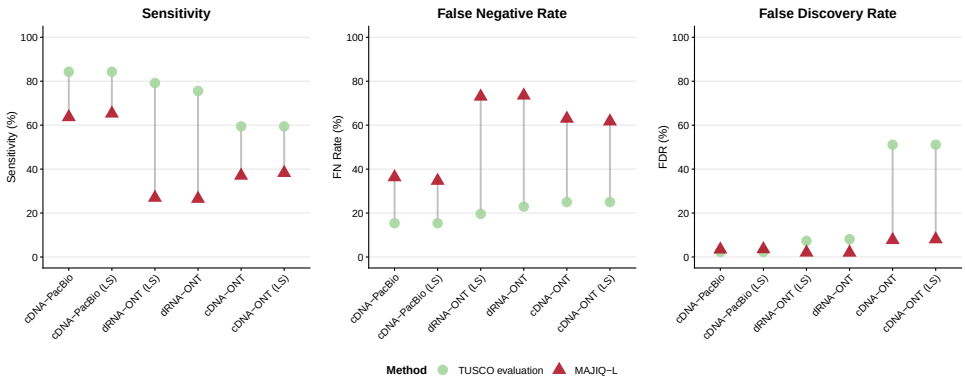
Supplementary Figure 2: Comparative properties of transcripts in the TUSCO gene set versus reference annotations in human and mouse. a–b, Transcript length distributions in discrete bins (<600 bp, 600–1kb, 1kb–2kb, 2kb–4kb, >4kb) for human **a**, and mouse **b**. **c–d**, Exon count per transcript shown as frequency within each dataset (%) for human **c**, and mouse **d**; the "10+" bin groups transcripts with ≥10 exons. **e–f**, Intron length distributions in discrete bins (<100, 100–1kb, 1kb–10kb, 10kb–100kb, >100kb) for human **e**, and mouse **f**. **g–h**, Density of exonic GC content (%) for human **g**, and mouse **h**; x-axis truncated to 20–80%.



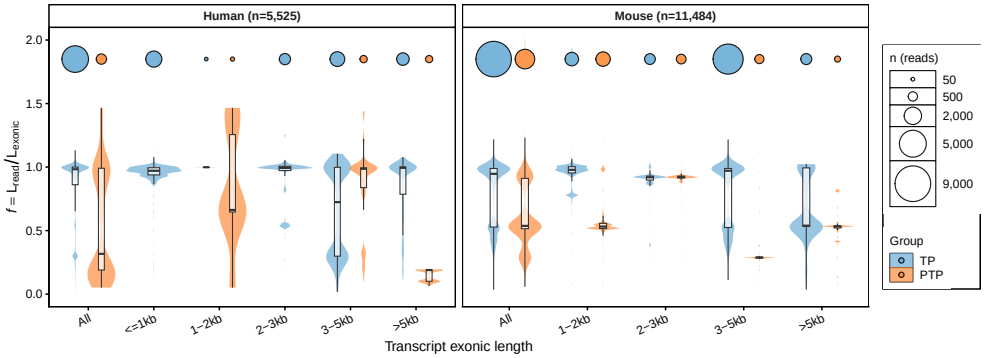
Supplementary Figure 3: Genes in the TUSCO gene set are highly expressed and display minimal novel splicing signal. Violin–box plots compare AlphaGenome-predicted scores for genes in the TUSCO gene set with single-isoform genes annotated in both GENCODE and RefSeq for human and mouse. **a**, Expression score (gene-level tissue-median RPKM). **b**, Splicing score (for each gene, the maximum AlphaGenome-predicted unannotated-to-annotated splice-junction count ratio across tissues). Values are plotted as $\log_{10}$ (score $+1 \times 10^{-4}$). Across both species, genes in the TUSCO gene set exhibit substantially higher expression and lower novel splicing score ratios than other single-isoform genes, consistent with their selection as widely expressed, splicing-invariant genes. Box centers mark medians; boxes span the interquartile range; whiskers extend 1.5× IQR. Colors: TUSCO (Human) light green, TUSCO (Mouse) dark green, Other (Human) light gray, Other (Mouse) dark gray.



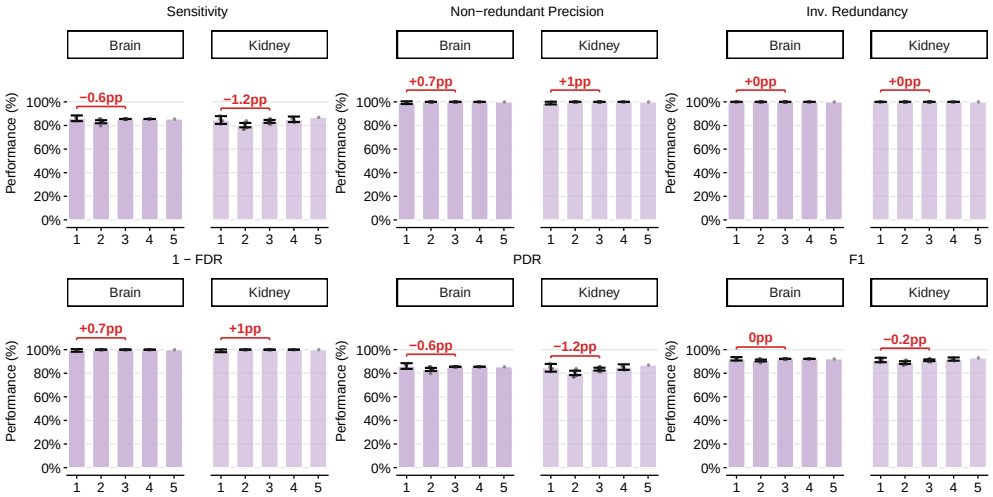
Supplementary Figure 4: False-negative rates differ primarily by sequencing platform and method. Raincloud plot showing false-negative rates from TUSCO evaluation across LRGASP datasets grouped by platform and library type (PacBio-cDNA, ONT-dRNA, ONT-cDNA). In this subset, FN rates cluster by platform/protocol, with PacBio-cDNA showing the lowest FN rates. Total genes in the TUSCO gene set: human $n = 46$; mouse $n = 32$.



Supplementary Figure 5: Comparison of TUSCO evaluation and MAJIQ-L benchmarking results. Left panel: Sensitivity (TP / (TP + FN)). Middle panel: False Negative rate (FN / (TP + FN)). Right panel: False Discovery Rate ($100 \times (N - TP)/N$). Metric definitions for MAJIQ-L are provided in Supplementary Table 2. Green circles represent TUSCO evaluation metrics; red triangles represent MAJIQ-L metrics. Lines connect the two methods for each pipeline. Only cDNA-PacBio consistently ranks best under both methods; dRNA-ONT and cDNA-ONT show divergent rankings that reflect different quality dimensions (RNA integrity vs. sequencing depth).



Supplementary Figure 6: Per-read coverage of transcripts in the TUSCO gene set in LRGASP Iso-Seq cDNA data, stratified by reference transcript exonic length. Violin-plus-box plots show the distribution of per-read coverage $f = L_{\text{read}}/L_{\text{exonic}}$ for reads mapped to the TUSCO gene set, separated by species (human WTC11, mouse ES) and binned by reference transcript exonic length (All, $\leq$1kb, 1–2kb, 2–3kb, 3–5kb, >5kb). Blue indicates TP (true positive, full-splice match) and orange indicates PTP (partial true positive). Only multi-exon transcripts in the TUSCO gene set are included; reference- and novel-evaluation classifications are combined; no TSS-based filtering; outliers retained; y-axis 0–2.0.



Supplementary Figure 7: SIRV spike-in controls exhibit minimal dependence on replicate number. Concordance metrics (sensitivity, non-redundant precision, inverse redundancy, 1-FDR, PDR, F1) were computed on FLAIR-identified SIRV transcripts from PacBio Sequel IIe cDNA libraries (∼5 M reads per library). Metrics were summarized across all replicate combinations for 1–5 biological replicates (bars: mean with 95% CI; points: individual combinations). Performance was essentially invariant with replicate number: precision, inverse redundancy and 1-FDR were ∼100% at all levels; brain sensitivity was 86.1% (83.7–88.5) at 1 replicate and ∼85.5% at 3–5; kidney sensitivity ranged 80.3% (78.4–82.1; 2 replicates) to 87.0% (5 replicates). F1 remained stable (brain ∼92.2%; kidney ∼89.0–93.0%). These results indicate that SIRV spike-ins yield highly consistent concordance irrespective of replicate count.

2 Supplementary Tables

Supplementary Table 1: Cosine similarity between TUSCO and SIRV benchmarks. Six evaluation metrics—non-redundant sensitivity (Sn), non-redundant precision (nrPre), inverse redundancy (1/red), one minus false-discovery rate (1-FDR), positive detection rate (PDR), and redundant precision (rPre)—were computed as percentages for SIRVs and TUSCO across pipelines spanning distinct data modalities (direct RNA or cDNA; ONT or PacBio). "LS" indicates integration of long- and short-read evidence. Values near 1 denote closely matched benchmark performance.

Cell type	Pipeline	Cosim
Mouse ES	ES dRNA ONT	0.9903
Mouse ES	ES cDNA ONT	0.9893
Mouse ES	ES cDNA PacBio	0.9992
Mouse ES	ES dRNA ONT LS	0.9903
Mouse ES	ES cDNA ONT LS	0.9882
Mouse ES	ES cDNA PacBio LS	0.9992
Human WTC11	WTC11 dRNA ONT	0.9966
Human WTC11	WTC11 cDNA ONT	0.9493
Human WTC11	WTC11 cDNA PacBio	0.9973
Human WTC11	WTC11 dRNA ONT LS	0.9977
Human WTC11	WTC11 cDNA ONT LS	0.9493
Human WTC11	WTC11 cDNA PacBio LS	0.9969

Supplementary Table 2: MAJIQ-L junction-level benchmarking results. Junction counts for each LRGASP library preparation workflow comparing long-read reconstructions against matched short-read MAJIQ data from WTC11 human samples. TP = True Positive (junction detected by both LR and MAJIQ); FP = False Positive (junction detected by LR but not MAJIQ); FN = False Negative (junction detected by MAJIQ but not LR). Known junctions are present in the reference annotation; novel junctions are absent. "LS" indicates integration of long- and short-read evidence.

Pipeline	TP-known	FP-known	FN-known	TP-novel	FP-novel	FN-novel
cDNA-ONT	80,021	4,397	124,611	839	2,385	12,847
cDNA-ONT (LS)	81,343	4,447	123,289	2,221	2,904	11,465
cDNA-PacBio	138,081	3,617	66,551	939	1,312	12,747
cDNA-PacBio (LS)	139,903	3,637	64,729	2,655	1,645	11,031
dRNA-ONT	57,801	971	146,831	52	208	13,634
dRNA-ONT (LS)	58,718	980	145,914	181	237	13,505

Supplementary Table 3: Performance metrics under native and TUSCO-novel annotations. Sensitivity (Sn) and non-redundant precision (nrPre) across library preparation protocols. Values show mean with range (min–max) across n protocol variants (CapTrap/cDNA × PacBio/ONT). Iso-Seq + SQANTI3 ML has $n = 1$ (PacBio cDNA only).

			Sn (%)		nrPre (%)	
Tool	Species	n	ref	novel	ref	novel
Bambu	Human	4	100.0	48.9 (39.1–73.9)	99.0 (95.8–100.0)	30.2 (25.0–41.5)
Bambu	Mouse	4	100.0	56.0 (48.0–72.0)	96.2	32.4 (30.0–37.5)
StringTie2	Human	4	70.7 (52.2–95.7)	39.1 (30.4–47.8)	69.5 (58.3–95.7)	37.1 (33.3–45.8)
StringTie2	Mouse	4	78.0 (68.0–92.0)	27.0 (24.0–32.0)	74.8 (65.5–92.0)	22.5 (17.6–26.1)
FLAIR	Human	4	68.5 (52.2–91.3)	42.4 (17.4–65.2)	77.4 (75.0–80.8)	47.4 (22.2–71.4)
FLAIR	Mouse	4	79.0 (68.0–96.0)	53.0 (44.0–68.0)	61.9 (50.0–68.6)	35.7 (24.5–50.0)
Iso-Seq + SQ3 ML	Human	1	52.2	52.2	57.1	63.2
Iso-Seq + SQ3 ML	Mouse	1	52.0	44.0	50.0	55.0

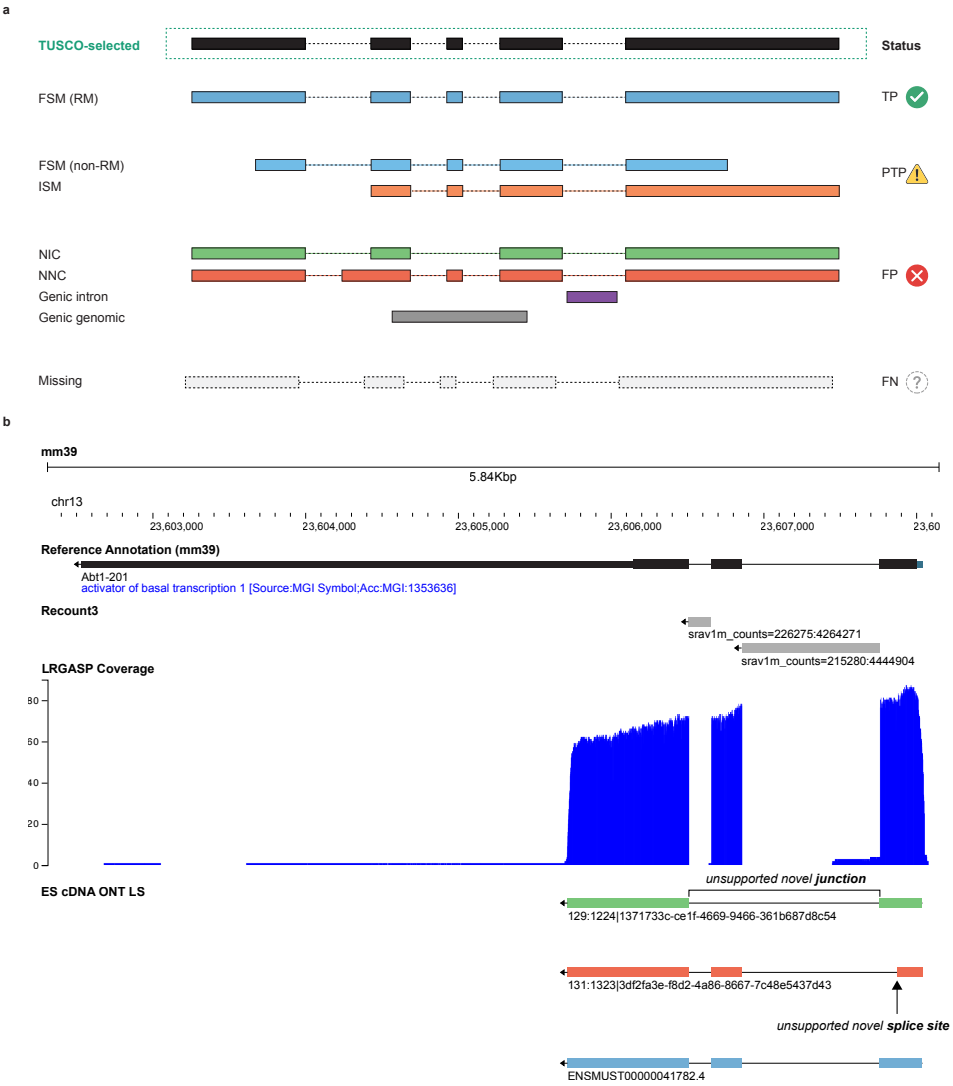
Supplementary Table 4: PacBio Iso-Seq read counts. Full-length non-chimeric (FLNC) HiFi read counts for PacBio Iso-Seq samples from mouse kidney (K31–K35) and brain (B31–B35). These per-sample yields provide the sequencing-depth context for the replicate analysis in Fig. 5 and Supplementary Fig. 6 and help readers assess the trade-off between additional replicates versus deeper sequencing under a fixed budget. Each value represents the total number of mapped reads per sample, as determined by samtools.

Sample	Read Count	Tissue
K31	4,593,279	Kidney
K32	5,136,502	Kidney
K33	3,902,779	Kidney
K34	5,436,609	Kidney
K35	5,096,149	Kidney
B31	3,684,709	Brain
B32	3,854,778	Brain
B33	4,413,078	Brain
B34	3,888,139	Brain
B35	3,914,079	Brain

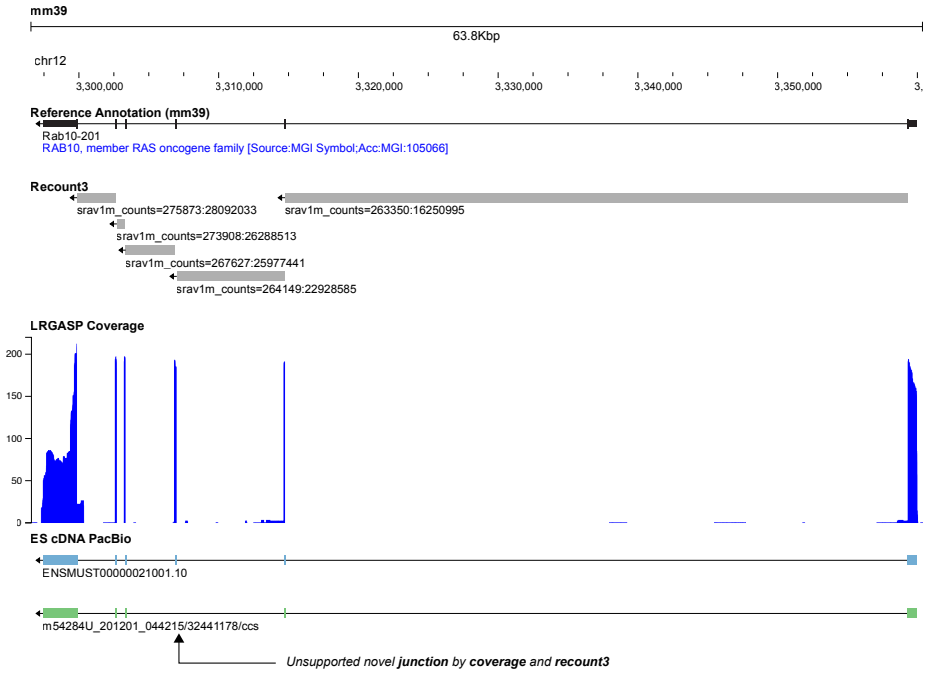
3 Supplementary Notes: TUSCO False Positive Validation

The TUSCO gene set provides a curated ground truth for benchmarking long-read transcriptome reconstruction. This supplementary note documents specific genes that were flagged as false positive (FP) transcript models within the TUSCO evaluation, demonstrating why the observed deviations should not be interpreted as novel isoforms.

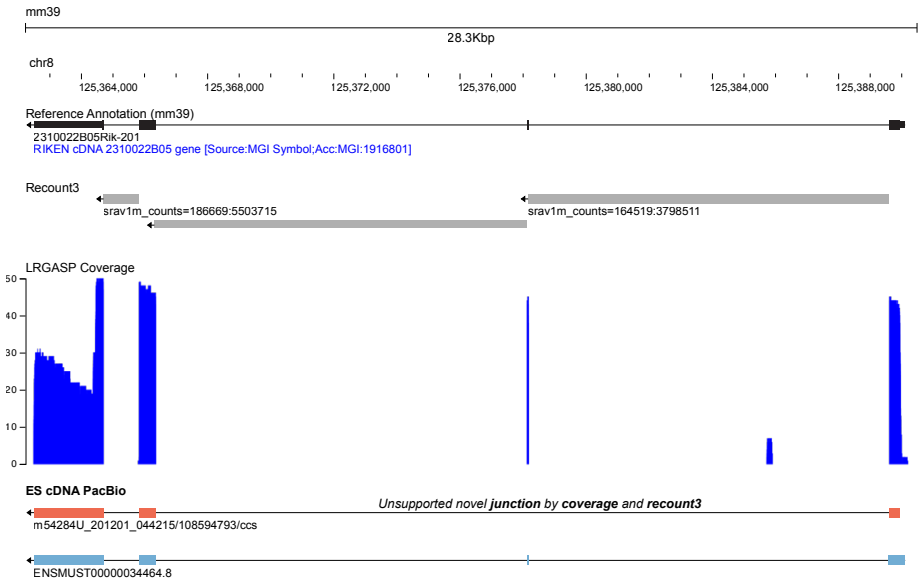
Each validation figure combines LRGASP coverage bigWig tracks (genome-wide per-base coverage merged from the dRNA-ONT, cDNA-ONT, CapTrap-ONT, cDNA-PacBio, and CapTrap-PacBio library protocols) alongside Recount3 short-read support (https://rna.recount.bio/) to contextualize the findings. These complementary datasets provide independent evidence to distinguish genuine biological variation from technical artifacts.



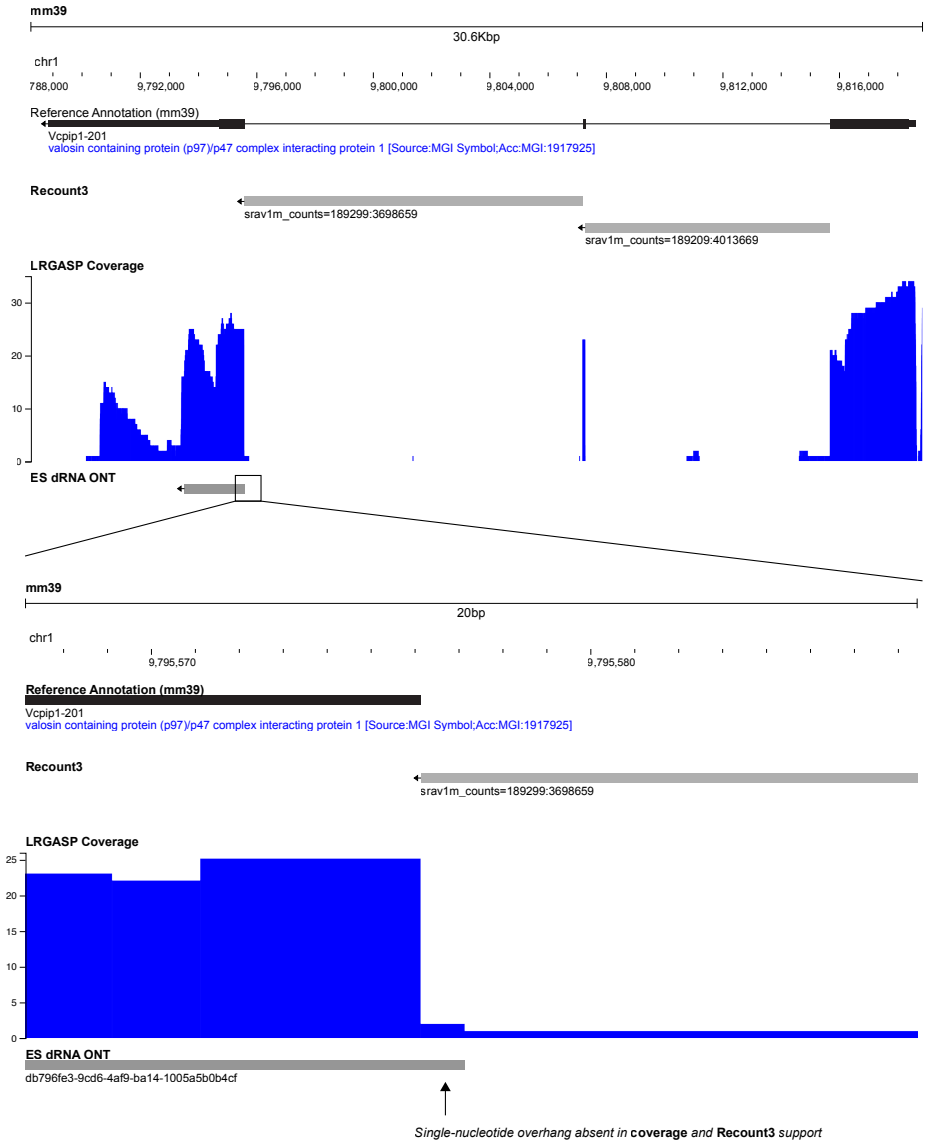
Supplementary Note 1: Validation of *Abt1* (ENSMUSG00000036376) as a false positive under TUSCO evaluation. a, Genomic view of the *Abt1* locus showing the reference transcript structure (blue) and the putative novel isoform detected by long-read sequencing (red). **b**, LRGASP coverage tracks demonstrate absence of reads supporting the proposed alternative splice junction. The merged coverage from multiple long-read protocols (dRNA-ONT, cDNA-ONT, CapTrap-ONT, cDNA-PacBio, CapTrap-PacBio) shows consistent splicing patterns matching the reference annotation. Recount3 short-read data corroborate this finding, with no evidence of the suggested splice variant. The apparent deviation represents a sequencing artifact rather than a genuine biological isoform, validating the false-positive classification under TUSCO evaluation.



Supplementary Note 2: Validation of *Rab10* (ENSMUSG00000020671) as a false positive under TUSCO evaluation. a, Structural comparison between the reference *Rab10* transcript and the candidate alternative isoform identified by transcript reconstruction algorithms. **b**, Comprehensive analysis of LRGASP coverage tracks reveals no supporting evidence for the proposed splice junction modification. The uniform coverage pattern across all long-read sequencing modalities aligns exclusively with the reference annotation. Short-read validation through Recount3 confirms the absence of reads supporting the putative variant junction. This evidence reinforces the classification of this detection as a technical artifact, demonstrating the utility of TUSCO evaluation in distinguishing genuine biological variation from reconstruction errors.



Supplementary Note 3: Validation of *2310022B05Rik* (ENSMUSG00000031983) as a false positive under TUSCO evaluation. a, Comparative view of the reference transcript architecture versus the algorithmically predicted variant for this uncharacterized gene. **b**, Multi-platform coverage analysis using LRGASP bigWig tracks shows no evidence supporting the suggested splice site alteration. The coverage profiles from direct RNA, cDNA, and CapTrap methodologies across both ONT and PacBio platforms consistently match the reference exon-intron structure. Independent validation using Recount3 short-read data corroborates the absence of the proposed junction, confirming that this represents a false-positive detection attributable to technical limitations rather than biological splicing variation.



Supplementary Note 4: Validation of *Vcpip1* (ENSMUSG00000045210) as a false positive under TUSCO evaluation. a, Detailed genomic view of the *Vcpip1* locus highlighting the discrepancy between reference annotation and the computationally inferred transcript variant. **b**, High-resolution coverage analysis demonstrates the absence of supporting reads for the proposed single-nucleotide overhang modification. LRGASP tracks from the comprehensive panel of long-read protocols show consistent alignment with the reference transcript boundary, with no evidence of the suggested structural alteration. Recount3 short-read validation provides additional confirmation, showing uniform coverage patterns that match the reference annotation. This analysis conclusively demonstrates that the detected variant represents a mapping or basecalling artifact rather than a genuine transcript isoform, supporting its classification as a false positive under TUSCO evaluation.